

Multiple Choice

1. **Answer:** B

Options: A) $O(1)$; B) $O(N)$; C) $O(\log N)$; D) $O(N^2)$

Note: Correct option: B - $O(N)$

2. **Answer:** D

Options: A) L 0 norm; B) L 1 norm; C) L 2 norm; D) either (a) or (b)

Note: Correct option: D - either (a) or (b)

3. **Answer:** A

Options: A) Lower variance; B) Higher variance; C) Same variance; D) None of the above

Note: Correct option: A - Lower variance

4. **Answer:** A

Options: A) random crop and horizontal flip; B) random crop and vertical flip; C) posterization; D) dithering

Note: Correct option: A - random crop and horizontal flip

5. **Answer:** B

Options: A) Ridge; B) Lasso; C) both (a) and (b); D) neither (a) nor (b)

Note: Correct option: B - Lasso

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'It discovers causal relationships'.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: ' $P(E, F)$, $P(H)$, $P(E, F|H)$ '.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'False, False'.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Detecting fraudulent credit card transactions'.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** `def find_first_occurrence(A, x): | return result`

Note: Students should provide a detailed explanation referencing algorithm steps.

12. **Answer:** `def min_of_two( x, y ): | return x | return y`

Note: Students should provide a detailed explanation referencing algorithm steps.

End of Answer Key